

Felipe "Yuee" Lemos

Security Engineer | Red Team & Blue Team | Cybersecurity Researcher | AI & Rust Systems
Rio de Janeiro, Brazil · <https://www.linkedin.com/in/yuee/> · <https://github.com/edtw> · NEXUS on Roblox

SUMMARY

Security engineer in Rio de Janeiro. I build Rust runtimes with local AI and break systems on purpose in an isolated lab. Background in backend, infrastructure, distributed systems, and live Roblox gameplay. OSCP in progress. Open to red team, blue team, research, and AI engineering roles.

TECHNICAL SKILLS

- **Backend & product:** TypeScript · Python · Node.js · FastAPI · React · Next.js · REST · GraphQL · Socket.IO
- **Systems & infrastructure:** Rust · C/C++ · Windows internals · Linux/macOS · Docker · CI/CD · PostgreSQL · Redis
- **Red team:** Adversary emulation · Windows internals · binary & protocol analysis · attack-surface review
- **Blue team:** Detection engineering · telemetry & logging · threat modeling · hardening validation
- **AI & local inference:** Candle · GGUF · TinyML · capability boundaries · PyTorch · OpenCV · Lua

EXPERIENCE

Roblox Studio Developer, NEXUS (Roblox Studio)

Current

Gameplay development for a live PvP arena experience. Contributing to NEXUS in Roblox Studio, an evolving combat experience centered on distinctive characters, abilities, visual effects, and strategic player interaction. The role brings together gameplay logic, rapid iteration, technical collaboration, and attention to how systems feel in play.

SELECTED PROJECTS

Karma Network · Spirix: RUST · DISTRIBUTED SYSTEMS · MESH / BLOCKCHAIN / COMPUTE

Mesh network framework in Rust built around Proof of Antiquity blockchain, distributed computing, and agent capabilities.

- Contribute to a large Rust workspace unifying blockchain, mesh networking, and distributed compute
- Work across DHT overlay, DAG state sync, relay services, and federated-learning primitives
- Integrate daily across site, API, agents, and docs in a fast-moving monorepo

Rust · Tokio · Axum · SQLx · Candle · DHT

Micro Runtime: RUST · SECURITY ENGINEERING · LOCAL AI · OT FLEET RUNTIME

A Rust runtime for small machines and industrial edge fleets. It runs local AI inside hard limits: a tiny INT8 model VM, GGUF inference on CPU, and a capability broker where every model decision is authorized before it reaches the host. Ships with a hardened fleet agent (mutual TLS, site PKI, signed extensions), an operator HMI, memory-safe workers, sealed weights, and CI gates that keep the architecture honest.

- Built a feature-gated Rust runtime where capabilities compile in or out by build profile without forking the codebase
- Added local AI execution: memory-mapped GGUF, a no-heap INT8 TinyML VM with hard budgets, and a capability broker that authorizes each model-proposed action
- Engineered a hardened fleet agent with mutual TLS, site PKI, signed extensions, and a closed HostOp ISA in isolated session workers

Rust · no_std · Candle · GGUF · TinyML · mTLS · PKI · Windows internals

ESTH: Enterprise Scripting & Telemetry Host: C++ · PYTHON · LICENSED SYSTEMS AGENT

Licensed client + agent + cloud API for scripting and telemetry against an authorized first-party game client in isolated lab VMs.

- Built a cross-platform systems agent with local IPC, queued scheduling, and data-driven config sync
- Shipped license/device binding with ownership-gated, encrypted configuration delivery
- Backed it with integration tests, snapshot diagnostics, and docs written for fail-closed behavior

C++ · CMake · FastAPI · AES-GCM · pytest

Marambaia PDV: FULL STACK · REAL TIME · OPERATIONS

Multi-surface restaurant operations platform covering products, tables, orders, cash registers, customer QR ordering, analytics, backups, access control, and real-time synchronization. Includes work on race conditions and order-total consistency.

- Built real-time ordering, QR customer flows, and cash/table management for a restaurant POS
- Hunted down race conditions and order-total bugs across concurrent admin and customer surfaces

React · Node.js · MongoDB · Socket.IO

Prisma Platform: AI SERVICES · AUTOMATION · PLATFORM ARCHITECTURE

Service-oriented platform for WhatsApp connectivity and AI-assisted workflows. Separates web application, messaging service, AI handling, provider abstraction with fallback, intent detection, analytics, persistence, caching, and containerized operations.

- Designed a multi-service WhatsApp/AI platform with provider fallback and session management
- Drew hard lines between web, messaging, AI, persistence, and cache behind Docker Compose

Next.js · FastAPI · PostgreSQL · Redis · Docker

Airshipper: RUST · RELEASE ENGINEERING · CROSS-PLATFORM

Customized desktop launcher and release service with platform artifact mapping, GitHub release integration, metrics, database-backed metadata, installers, and multi-platform build pipelines. Based on the open-source Veloren Airshipper; customization only, with upstream attribution.

- Customized the open-source Veloren launcher for GitHub-hosted release channels
- Mapped platform artifacts, metrics, and metadata with automated multi-OS packaging

Rust · Iced · Tokio · SQLx · CI

SECURITY RESEARCH

Offense and defense, always in an isolated lab or authorized scope. I emulate adversaries to learn how systems break, then turn that into detections and hardening. Sensitive work stays private; this page stays at architecture level.

- Red team: adversary emulation, Windows internals, binary and protocol analysis in isolated labs
- Blue team: detection engineering, telemetry and logging, threat modeling, hardening validation
- Research: cryptography, runtime integrity, supply chain, and security boundaries for local AI
- Practice: authorization first, reproducible evidence, and defensive outcomes written down

CERTIFICATIONS

- OSCP (Offensive Security Certified Professional), in progress through self-study and lab practice
- TryHackMe, active learner across red-team and blue-team tracks

LANGUAGES & INTERESTS

Portuguese (native) · English (professional working proficiency)
Red-team and blue-team home lab · CTFs · local AI and open-source Rust · game development (Roblox Studio)

NOTE

Sensitive repositories are described as private case studies without operational code. Architecture detail is available in technical interviews when appropriate. Updated 2026-09-10 · v2.1.0